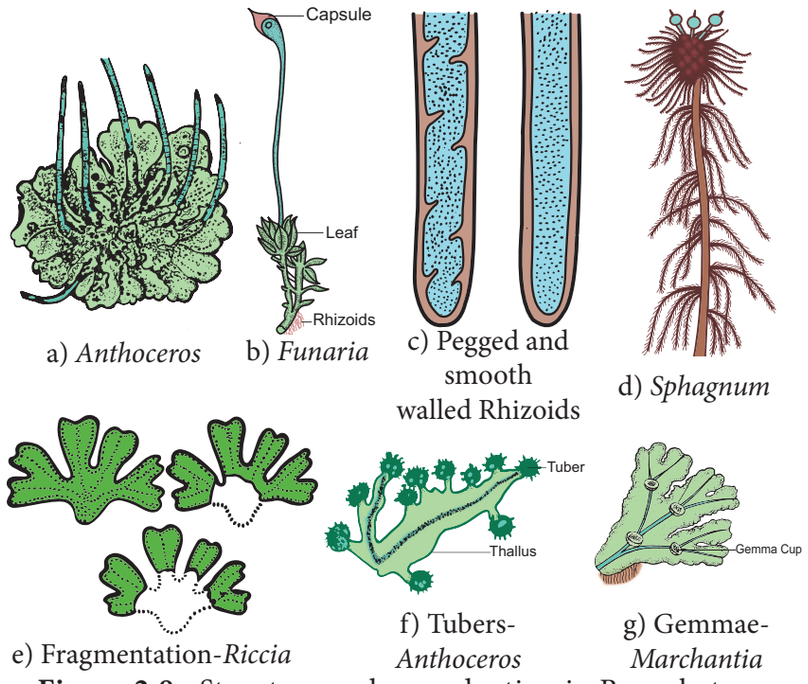


a) *Anthoceros* b) *Funaria* c) Pegged and smooth walled Rhizoids d) *Sphagnum*

e) Fragmentation-*Riccia* f) Tubers-*Anthoceros* g) Gemmae-*Marchantia*

Figure 2.8: Structure and reproduction in Bryophytes

develops from the outer cell). The embryo divides and give rise to sporophyte.

- The sporophyte is dependent on gametophyte.
- It is differentiated into three recognizable parts namely foot, seta and capsule.
- Foot is the basal portion and is embedded in the gametophyte through which water and nutrients are supplied for the sporophyte. The diploid spore mother cells found in the capsule region undergoes meiotic division and give rise to haploid spores. Bryophytes are homosporous. In some sporophytes elaters are present and help in dispersal of spores (Example: *Marchantia*). The spores germinate to produce gametophyte.
- The zygote, embryo and the sporogonium constitute sporophytic phase. The green long living haploid phase is called gametophytic phase. The haploid gametophytic phase alternates with diploid sporophyte and shows heterologous alternation of generation.

2.4.2 Classification of Bryophytes

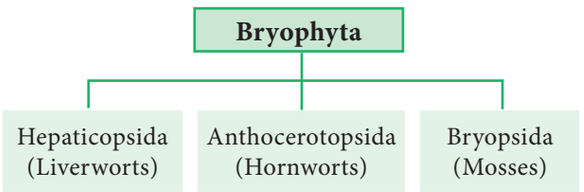
Proskauer in the year 1957 classified Bryophytes into 3 Classes namely

i. **Hepaticopsida** (*Riccia, Marchantia, Porella* and *Riella*)

ii. **Anthocerotopsida** (*Anthoceros* and *Dendroceros*)

iii. **Bryopsida** (*Funaria, Polytrichum* and *Sphagnum*).

The outline of the classification is given below



Class: Hepaticopsida

They are lower forms of Bryophytes. They are more simple in structure than mosses and more confined to damp and shady places. They have an undifferentiated thallus. Protonernal stage is absent. Sporophyte is very simple and short lived. In some, the foot and seta are absent. Example *Riccia*.

Class: Anthocerotopsida

Gametophyte is an undifferentiated thallus. Rhizoids are unicellular and unbranched. Protonemal stage is absent. Sporophyte is differentiated into foot and capsule and seta is absent Example: *Anthoceros*.

Class: Bryopsida

These are higher forms in which the gametophyte is differentiated into 'stem' like and 'leaf' like parts and the former showing radial symmetry. Rhizoids are multi-cellular and branched. Protonemal stage is present. Sporophyte is differentiated into foot, seta

XIth_Botany_Chapter_2_EM.indd 42 5/17/2020 9:01:05 PM